Clustering: Unsupervised Learning

1 k-medoid clustering

Clustering groups similar data points without labels. Unlike k-means, where centroids may not be actual observations, k-medoids chooses *medoids*—actual data points—as cluster centers, improving robustness to outliers.

2 Partitioning Around Medoids (PAM) Algorithm

Given n data points and desired k clusters:

1. **Initialize:** Randomly select k points as initial medoids.
2. **Assignment:** Assign each point to the nearest medoid (using Euclidean distance).
3. **Update:** For each medoid m and each non-medoid point o, swap m with o and compute total clustering cost. If cost decreases, keep the swap.
4. **Repeat:** Iterate Assignment and Update until no swap reduces cost.

3 Numerical Example

Consider six 2D points and $k = 2$:

Point	x	y
P_1	2	6
P_2	3	4
P_3	3	8
P_4	5	7
P_5	6	2
P_6	7	3

3.1 Step 1: Initialization

Choose initial medoids:

$$M_1 = P_1(2, 6), \quad M_2 = P_5(6, 2).$$

3.2 Step 2: First Assignment

Compute distances to M_1 and M_2, then assign:

Point	$d(\cdot, M_1)$	$d(\cdot, M_2)$	Cluster
P_1	0	5.66	1
P_2	2.24	3.61	1
P_3	2.24	6.71	1
P_4	3.16	5.10	1
P_5	5.66	0	2
P_6	5.83	1.41	2

Total cost =

$$(0 + 2.24 + 2.24 + 3.16) + (0 + 1.41) = 10.46.$$

3.3 Step 3: Evaluate Potential Swaps

Test swapping M_1 with each non-medoid P_2, P_3, P_4, P_6 (keeping $M_2 = P_5$) and compute the resulting cost:

Swap	Total Cost
$M_1 \leftrightarrow P_2$	11.26
$M_1 \leftrightarrow P_3$	**9.50**
$M_1 \leftrightarrow P_4$	10.42
$M_1 \leftrightarrow P_6$	20.14

Best swap reduces cost from 10.46 to 9.50 by replacing M_1 with P_3.

Additionally, to ensure completeness, evaluate swapping the second medoid $M_2 = P_5$ with each non-medoid P_2, P_3, P_4, P_6 while keeping $M_1 = P_1$. None of these swaps decrease the total cost (e.g., swapping $M_2 \leftrightarrow P_2$ yields cost 13.13), so M_2 remains P_5.

3.4 Step 4: Update Medoids

Update medoids:

$$M_1 = P_3(3, 8), \quad M_2 = P_5(6, 2).$$

3.5 Step 5: Second Assignment

Recompute distances and assign clusters:

Point	$d(\cdot, P_3)$	$d(\cdot, P_5)$	Cluster
P_1	2.24	5.66	1
P_2	4.00	3.61	2
P_3	0	6.71	1
P_4	2.24	5.10	1
P_5	6.71	0	2
P_6	6.40	1.41	2

Total cost $= (2.24 + 0 + 2.24) + (3.61 + 0 + 1.41) = 9.50$.

3.6 Step 6: Termination

No further swap reduces the cost, so the algorithm terminates with final medoids and clusters:

- Medoids: $M_1 = P_3(3, 8)$, $M_2 = P_5(6, 2)$
- Cluster 1: $\{P_1, P_3, P_4\}$
- Cluster 2: $\{P_2, P_5, P_6\}$

4 Conclusion

K-medoids clustering provides interpretable centers by selecting actual points as medoids and is robust to outliers. Though swap evaluation incurs higher computation than k-means, it yields meaningful cluster representatives.

K-Means Clustering: Definition and Objective

K-means clustering partitions a set of n data points $\{x_i\}_{i=1}^n$ in $\mathbb{R}^d$ into k clusters so as to minimize the within-cluster sum of squares. The objective function is

$$J \;=\; \sum_{i=1}^{n}\sum_{j=1}^{k} r_{ij}\,\|x_i-\mu_j\|^2,$$

where

$$r_{ij}=\begin{cases}1, & \text{if } x_i \text{ is assigned to cluster } j,\\ 0, & \text{otherwise,}\end{cases} \qquad \mu_j=\frac{\sum_{i=1}^n r_{ij}\,x_i}{\sum_{i=1}^n r_{ij}}$$

is the centroid of cluster j.

Algorithmic Steps

1. **Initialize:** choose initial centroids $\{\mu_j^{(0)}\}_{j=1}^k$.
2. **Assignment:** for each i, set
$$r_{ij}=\begin{cases}1, & j=\arg\min_\ell \|x_i-\mu_\ell^{(t)}\|^2,\\ 0, & \text{otherwise.}\end{cases}$$
3. **Update:** recompute each centroid
$$\mu_j^{(t+1)}=\frac{\sum_{i=1}^n r_{ij}\,x_i}{\sum_{i=1}^n r_{ij}}.$$
4. **Repeat** steps 2–3 until assignments no longer change or J converges.

Numerical Example

Data: six points in $\mathbb{R}^2$,

$$\{\,P_1=(1,2),\,P_2=(1,4),\,P_3=(2,3),\,P_4=(5,8),\,P_5=(7,8),\,P_6=(8,7)\}.$$

Set $k=2$, with initial centroids

$$\mu_1^{(0)}=P_1=(1,2), \qquad \mu_2^{(0)}=P_4=(5,8).$$

Iteration 0: Assignment & Cost

Point	$d^2(x_i,\mu_1^{(0)})$	$d^2(x_i,\mu_2^{(0)})$	Assigned cluster
$P_1=(1,2)$	0	52	1
$P_2=(1,4)$	4	32	1
$P_3=(2,3)$	2	41	1
$P_4=(5,8)$	61	0	2
$P_5=(7,8)$	85	4	2
$P_6=(8,7)$	74	10	2

Thus

$$J^{(0)}=(0+4+2)\;+\;(0+4+10)\;=\;6+14=20.$$

Iteration 0: Update Centroids

$$\mu_1^{(1)}=\frac{(1,2)+(1,4)+(2,3)}{3}=\Big(\tfrac{4}{3},\,3\Big), \qquad \mu_2^{(1)}=\frac{(5,8)+(7,8)+(8,7)}{3}=\Big(\tfrac{20}{3},\,\tfrac{23}{3}\Big).$$

Iteration 1: Assignment & Cost

Re-compute distances to $\mu_1^{(1)} \approx (1.33, 3)$ and $\mu_2^{(1)} \approx (6.67, 7.67)$:

Point	$d^2(x_i, \mu_1^{(1)})$	$d^2(x_i, \mu_2^{(1)})$	Assigned cluster
P_1	1.11	64.22	1
P_2	1.11	42.89	1
P_3	0.44	50.22	1
P_4	38.44	2.89	2
P_5	70.22	0.22	2
P_6	45.78	2.22	2

Hence

$$J^{(1)} = (1.11 + 1.11 + 0.44) \; + \; (2.89 + 0.22 + 2.22) \; = \; 2.66 + 5.33 = 8.00.$$

Since assignments did not change from Iteration 0, the algorithm converges here.

Conclusion

- The cost J decreased from 20 to 8, demonstrating convergence.
- Final clusters:
$$C_1 = \{P_1, P_2, P_3\}, \quad C_2 = \{P_4, P_5, P_6\}.$$
- Final centroids:
$$\mu_1 = \left(\tfrac{4}{3}, 3\right), \quad \mu_2 = \left(\tfrac{20}{3}, \tfrac{23}{3}\right).$$

Hierarchical Clustering: Numerical Illustration

Hierarchical (agglomerative) clustering builds a hierarchy of clusters by starting with each point in its own cluster and then repeatedly merging the two closest clusters according to a linkage criterion.

Dataset

Consider the one-dimensional dataset

$$X = \{2,\ 5,\ 8,\ 12\}.$$

Step 1: Pairwise Distance Matrix

Compute all pairwise distances $d(x_i, x_j) = |x_i - x_j|$:

	2	5	8	12
2	0	3	6	10
5	3	0	3	7
8	6	3	0	4
12	10	7	4	0

Table 1: Initial pairwise distances.

Cost Function

For any cluster C, define the within-cluster sum-of-squares

$$\text{cost}(C) \ = \ \sum_{x \in C} (x - \mu_C)^2, \quad \mu_C = \frac{1}{|C|} \sum_{x \in C} x.$$

The total cost at each stage is

$$J \ = \ \sum_{\text{clusters } C} \text{cost}(C).$$

Agglomeration Steps

We use *single-linkage* (merge the two clusters with smallest minimum inter-point distance). At each merge we record the linkage distance and the updated total cost.

Step	Merged Clusters	Linkage d	Resulting Clusters	Total Cost J
0	—	—	$\{2\}, \{5\}, \{8\}, \{12\}$	0
1	$(2, 5)$	3	$\{2, 5\}, \{8\}, \{12\}$	4.5
2	$(\{2, 5\}, 8)$	3	$\{2, 5, 8\}, \{12\}$	18
3	$(\{2, 5, 8\}, 12)$	4	$\{2, 5, 8, 12\}$	54.75

Table 2: Merge sequence with distances and cost analysis.

Final Cluster Assignments

If we cut the tree at two clusters (just before Step 3), we obtain:

$$\text{Cluster 1: } \{2, 5, 8\}, \quad \text{Cluster 2: } \{12\}.$$

Brief Hierarchical Clustering Example

Dataset

$$X = \{2,\ 5,\ 8,\ 12\}.$$

Distance Matrix

$$D_{ij} = |x_i - x_j| \quad \Longrightarrow \quad D = \begin{pmatrix} 0 & 3 & 6 & 10 \\ 3 & 0 & 3 & 7 \\ 6 & 3 & 0 & 4 \\ 10 & 7 & 4 & 0 \end{pmatrix}.$$

Merge Sequence (Single Linkage)

- Step 1: Merge 2 and 5 (distance 3) $\Rightarrow$ $\{2,5\},\{8\},\{12\}$
- Step 2: Merge {2,5} and 8 (distance 3) $\Rightarrow$ $\{2,5,8\},\{12\}$
- Step 3: Merge {2,5,8} and 12 (distance 4) $\Rightarrow$ single cluster

Cutting the dendrogram after Step 2 yields two clusters:

$$\text{Cluster 1} = \{2,5,8\}, \quad \text{Cluster 2} = \{12\}.$$

Cost Function For any cluster C with mean μ_C,

$$\text{cost}(C) \;=\; \sum_{x \in C} (x - \mu_C)^2.$$

Here,

$$\mu_{\{2,5,8\}} = \frac{2+5+8}{3} = 5, \quad \text{cost}(\{2,5,8\}) = (2-5)^2 + (5-5)^2 + (8-5)^2 = 18,$$

$$\mu_{\{12\}} = 12, \quad \text{cost}(\{12\}) = (12-12)^2 = 0.$$

Total cost for the two-cluster cut is $18 + 0 = 18$.

Dendrogram of Hierarchical Clustering

